

AI Gesture Model Trainer Report

Project: Hand Gesture Drone Control System | Generated: 2026-07-28 17:31:32

1. Dataset Summary

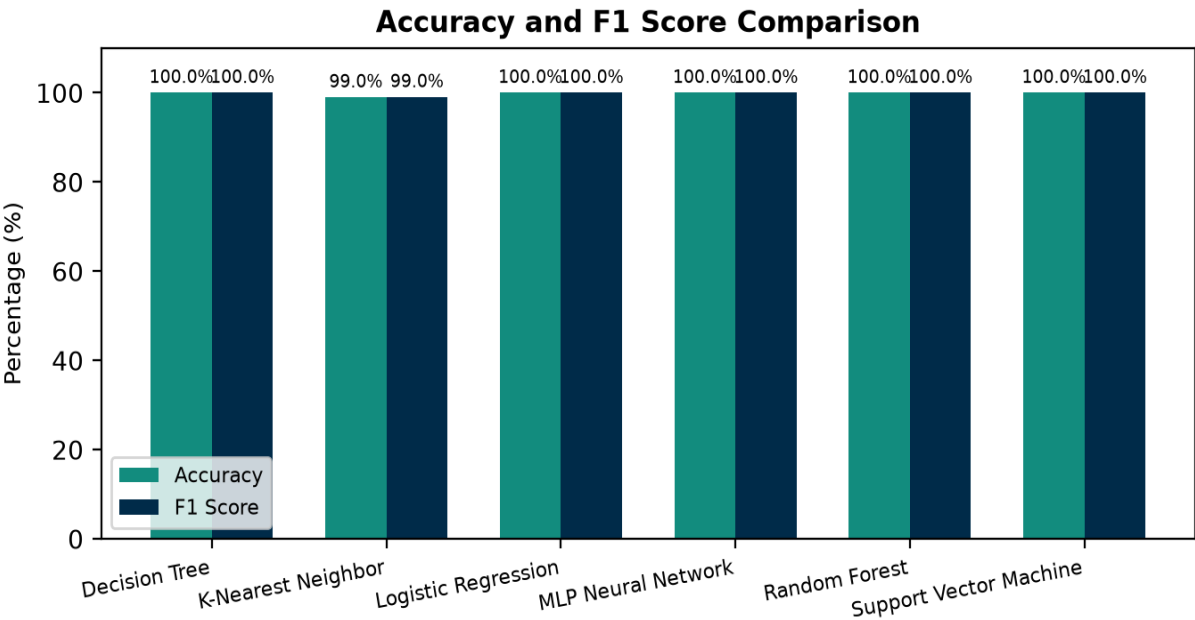
Metric	Value
Dataset Path	D:\university\SIC\AI\project_v2\dataset\gesture_dataset.csv
Total Sample Count	988
Feature Columns (Coordinates)	63
Number of Gesture Classes	9
Missing / NaN Values	0

2. Training Settings

Parameter	Value
Train / Test Ratio	80/20
Random State Seed	42
Cross Validation	5 Folds
Normalize Features	Yes (StandardScaler)
Shuffle Dataset	Yes

3. Model Performance Comparison

Model	Accuracy	Precision	Recall	F1	Train Time	Mem Size
Decision Tree	100.00%	100.00%	100.00%	100.00%	0.0169s	4.5 KB
K-Nearest Neighbor	98.99%	99.18%	98.94%	99.03%	0.0014s	395.8 KB
Logistic Regression	100.00%	100.00%	100.00%	100.00%	0.0171s	5.2 KB
MLP Neural Network	100.00%	100.00%	100.00%	100.00%	0.3487s	541.8 KB
Random Forest	100.00%	100.00%	100.00%	100.00%	0.2063s	1109.8 KB
Support Vector Machine	100.00%	100.00%	100.00%	100.00%	0.0792s	191.4 KB



4. Selected Best Model Details: Logistic Regression

Metric	Result Score
Accuracy Score	100.0000%
Precision (Macro)	100.0000%
Recall (Macro)	100.0000%
F1 Score (Macro)	100.0000%
Single Sample Latency	0.0378 ms
Model Serialization Size	5.24 KB

Classification Report Detail:

	precision	recall	f1-score	support
Backward	1.00	1.00	1.00	25
Down	1.00	1.00	1.00	22
Forward	1.00	1.00	1.00	21
Hover	1.00	1.00	1.00	22
Land	1.00	1.00	1.00	21
Left	1.00	1.00	1.00	22
Right	1.00	1.00	1.00	22
Takeoff	1.00	1.00	1.00	22
Up	1.00	1.00	1.00	21
accuracy			1.00	198
macro avg	1.00	1.00	1.00	198
weighted avg	1.00	1.00	1.00	198